2.4 Naming and discovery

In 2004, U.S. Senator Ted Kennedy was repeatedly stopped and subjected to additional screenings at airports. In at least one incident, he was barred from boarding his flight altogether. The problem? His name matched an entry in the U.S. Transportation Security Administration's (TSA's) No-Fly List, a list of people who are barred from air travel inside or to and from the United States. At the time, the No-Fly list relied heavily on names alone, rather than using unique identifiers like passport numbers. Because the TSA's system lacked a robust naming scheme and didn't handle discovery well, Senator Kennedy was treated as if he were the banned individual.

Naming is one of the most fundamental parts of digital identity and is notoriously difficult to get right. As the TSA example shows, getting it wrong can have far-reaching consequences. In the case of Senator Kennedy, he had a pretty good platform—the U.S. Senate—to fix the problem. Most people aren't so lucky.

An important lesson we can learn from this situation is that naming problems often lead to authorization problems. The failure of the No-Fly list to uniquely identify the correct Ted Kennedy led to an incorrect denial of service—a fundamental authorization failure caused by poor naming and discovery processes.

2.4.1 The role of names in digital identity

Names are labels that we give to things so that we don't have to continually describe them to each other. In that sense, they're a kind of shorthand for a person, place, or thing. Names are often not unique, and we interpret them depending on the context in which we see them.

We use the word *identifier* for names that are unique within a certain context, which we call a *namespace*. Social security numbers, usernames, email addresses, and database table names are all examples of identifiers. Digital identity systems need identifiers to recognize and remember the parties in a relationship. When you type your username in a login form, you're reminding the system on the other end who you are.